Deep Inelastic Scattering Revisited

Felix Hekhorn
University of Jyväskylä

Lecture notes for the Jyväskylä Summer School 2026

Abstract

Discuss Yadism [1].

Contents

Chapter 1

The basics

1.1 Introduction

1.1.1 Motivation and non-goals

We aim in this course to make a meaningful comparison between the theoretical predictions and the experimental measurement of a fully inclusive Deeply Inelastic Scattering (DIS) cross section. Instead of an abstract, theoretical discussion about DIS, which can be found in a standard lecture on particle physics, we're looking here to make a connection with real-life physics solving current research questions. The main focus is be naturally on PDF, for which DIS plays a major role. In fact, DIS was, is, and will be a very important constrain in determining Parton Distribution Functions (PDFs) - only recently, e.g. starting from NNPDF4.0 [2], LHC data starts to dominate in PDF fits. Stressing the importance and relevance of PDFs for any kind of perturbative QCD (pQCD) application and to understand the inner structure of hadrons is left to the literature [3–5]. In turn, DIS itself can be used to answer unresolved questions about PDFs, e.g. whether the proton has an intrinsic charm component [6].

In order to obtain a more realistic DIS treatment, we review general pQCD features and exemplify them using DIS. In fact, fully inclusive DIS comes in very handy, since here the effects mostly appear in their simplest form and when considering hadronic collisions, e.g. at the LHC, the strategies are mostly the same just more involved. We keep a phenomenological point of view with a specific focus on the numerical implementation as, eventually, we want to compare to the experimental measurements. We refrain from giving detailed proofs in this course and point to the standard textbooks [7–11] for that.

Instead, we consider different features as given and focus on their practical implications. However, a major complication in this course is the interconnection between almost all features and while we discuss them here in turn, we highlight also how they are intertwined with each other.

In recent years (and much more so hopefully in the coming years), DIS has become a major research topic again thanks to the advent of the Electron Ion Collider (EIC) [12]. While the planned physics program of the EIC goes far beyond standard fully inclusive DIS, this simplest case can serve as an introduction into more complicated topics.

The structure of this course follows in large parts the implementation of Yadism (Yet Another DIS module) [1, 13], which is developed and used by the NNPDF collaboration in their PDF determinations. The online documentation, which is available at

`https://yadism.readthedocs.io/` ,

may serve as additional, more technical introduction into the topic. Note that Yadism is an open-source project to which everybody can contribute[1].

Since DIS is a very wide field, it is worth spelling out explicitly some non-goals:

- We only consider pQCD in the collinear framework here. However, because DIS is such a powerful experiment, which offers a clean access to the inner structure of the protons it can be studied using other frameworks as well, such as Color Glass Condensates [14, 15] → ask H. Mäntysaari or T. Lappi
- We only consider fully inclusive cross sections here (to be defined in Section 1.2 and with a tiny exception to be discussed in Section 2.5). In particular, we do not consider any more complicated final states, such as, e.g., single jet or di-jet production [16].
- We only consider pQCD in the collinear framework here. While, Monte Carlo Event generators, such as, e.g., Pythia [17], are rooted in collinear pQCD they eventually go much beyond and we do not consider their details here [18] → ask I. Helenius.
- We only consider DIS here. In particular, we do not consider photo-production or elastic or diffractive cross sections. We define the necessary kinematics for the distinction below in Section 1.1.3.

[1] e.g. if some documentation can be improved

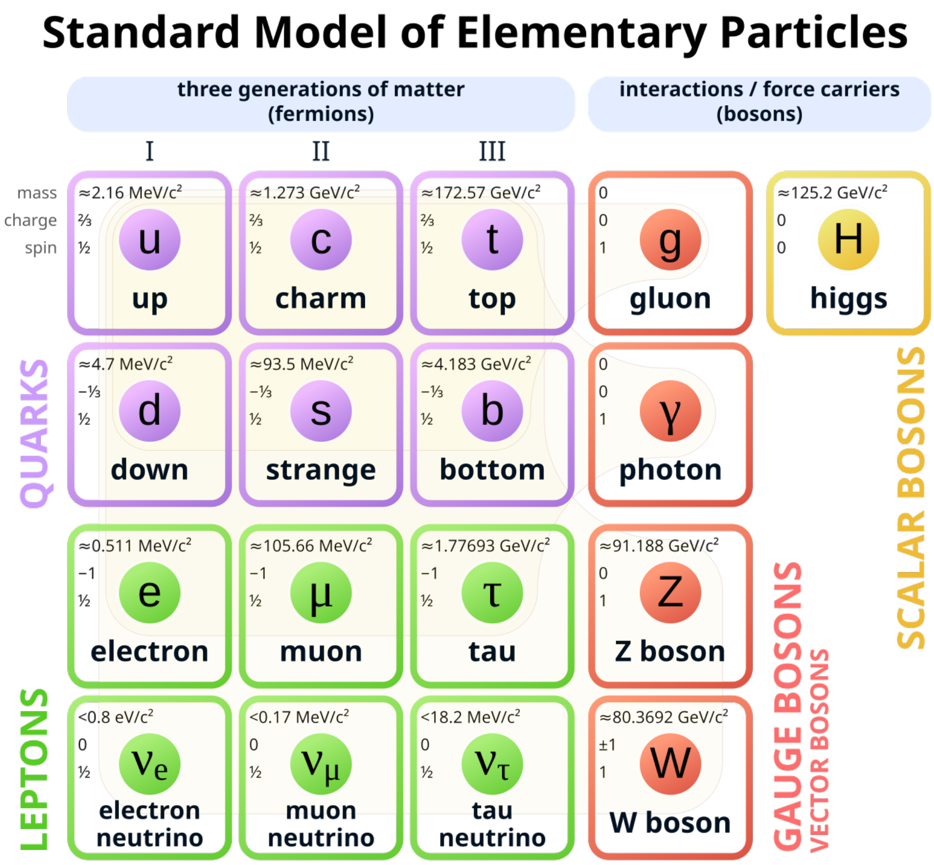

Figure 1.1: The Standard Model of Elementary Particles - taken from [19]

- We compute the theoretical predictions here. So while we try to keep the connection to the experimental side close, we do not discuss any specific details about how to measure cross sections.

1.1.2 Requirements

We need some basic knowledge from particle physics and, in particular, you need to be familiar with the Standard Model overview shown in Fig. 1.1. We discuss in this course each and everyone of these particles, with the exception of the three heaviest in each category: the heavies quark, the top, the heaviest lepton, the tau, and the heaviest boson, the Higgs. They work, in principle, mostly like their lighter brothers and sisters, but in practice are basically always neglected. We give a more thorough justification to discard them when we get to the actual numbers. All other particles explicitly appear in this course and you need to know their charges - see also Fig. 1.1.

Conclusion 1.1: Summary forces, particles, and charges

- The carrier of the strong force is the gluon
- Quarks have a strong charge, leptons don't
- The carriers of the electro-weak force are the photon, the Z boson, and the W boson
- Quarks and charged leptons have electro-magnetic charges
- Quarks and leptons have weak charges

We also need some basic knowledge from Quantum Field Theory (QFT).

1.1.3 The very basics

Let's start from the top:

Conclusion 1.2: DIS in words

fully inclusive DIS = fully inclusive deeply inelastic scattering

What does that actually mean? We begin from the back:

- "scattering" means, of course, we are scattering two particles with each other. In this particular case, we are scattering a lepton off a hadron and to make things simple in the beginning we consider an electron is scattered off a proton.
- "inelastic" means we are breaking the target, the proton, apart; it shatters into many pieces.
- "deeply" is an adverb[2] to inelastic and it means we are looking "deep" into the inside of the proton. And in order to resolve small things, as usual, we need an highly energetic probe.
- "fully inclusive" means we are observing a single particle in the final state, which is the electron, but nothing else.

Thus, we can write the same information again with a formula

[2] thus it is "deeply inelastic" or "deep-inelastic", but not a list of adjectives "deep inelastic"

Conclusion 1.3: DIS in a formula

fully inclusive DIS $= e^-(k) + p(P) \to e^-(k') + X$ with $-(k-k')^2 \gg \Lambda^2_{\mathrm{QCD}}$

where we have assigned the four-momenta k, P, k' to the incoming electron, the incoming proton and the outgoing electron respectively. As usual X marks the inclusiveness, i.e. we don't observe anything else apart from the electron. By requiring a large momentum-transfer between the incoming and the outgoing electron, i.e. their difference four-momentum vector has a large magnitude, we probe the proton with highly energetic parton (which we can think of as photon).

Next, we need a way to actually compute theory predictions for the cross sections, which you can measure in an experiment. We use the factorization theorem [20],

Conclusion 1.4: Factorization Theorem

$$\sigma(x, Q^2) = \sum_j \int_x^1 \frac{\mathrm{d}z}{z} C_j(z, Q^2) f_j(x/z, Q^2) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right) \tag{1.1}$$

$$= \sum_j \left(C_j(Q^2) \otimes f_j(Q^2)\right)(x) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right) \tag{1.2}$$

$$= \left(\mathbf{C}(Q^2) \otimes \mathbf{f}(Q^2)\right)(x) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right), \tag{1.3}$$

which states how to compute a cross section $\sigma(x, Q^2)$ from coefficient function $\mathbf{C}(z, Q^2)$, computable in pQCD, and a universal PDF $\mathbf{f}(\xi, Q^2)$. In the remaining part of the course we discuss, enhance and correct this formula in many different ways and basically challenge every aspect. We define each and every variable and function in Section 1.2, so we can focus for the moment on two other important aspects: universality and notation.

The crucial, physical aspect about Eq. (1.1) is universality: for any cross section σ measured at any x and Q^2 there is exactly one PDF $\mathbf{f}(\xi, Q^2)$. Instead, the coefficient functions $\mathbf{C}(z, Q^2)$ may depend on the definition of the cross section, but they are computable from first principles. This implies the predictive powers of our framework: if $\mathbf{C}$ is fixed and $\mathbf{f}$ is universal, we can measure σ in one set of experiments and then predict the outcome of another set of measurements σ'.

The other, more practical aspect of Eqs. (1.1) to (1.3) is the short-hand notation we introduce there. While Eq. (1.1) is the most canonical form of the factorization theorem, which can be found like this in most textbooks, we introduce in Eq. (1.2) the multiplicative convolution symbol $\otimes$. Note that we have suppressed the variable over which the convolution is performed and recall that convolution is associative, i.e. $(g \otimes h)(x) = (h \otimes g)(x)$. Finally, in Eq. (1.3) we introduce the vector notation, where we denote object, which have a non-trivial dependence on the parton flavor j, with bold font. Equation (1.3) looks nice and compact, but it worth recalling that it already hides a lot of details: specifically, $C_j(z, Q^2)$ is an object which depends on three different variables. We use vector notation and the index notation interchangeably depending on the context.

To Leading Order (LO) accuracy, the coefficient functions $\mathbf{C}$ greatly simplify and we find

$$\sigma(x, Q^2) = \sum_j e_{q_j}^2 f_j(x, Q^2)\,, \tag{1.4}$$

i.e. the cross section is given by the sum of the quark PDFs weighted by their electrical charges squared. Note that here the variables, which are used for the experimentally measured cross section on the left-hand-side, are also used as arguments of the PDFs on the right-hand-side. This is a pure coincidence only valid in the simplest case at LO and two historical comments can be made here. First, when studying older literature this equivalence can lead to confusion as, e.g., PDFs have been also called "structure functions" at the time, or what exactly we mean when talking about a general variable "x". Second, in the so-called "parton model", which was popular before QCD was invented, the index j just referred to a "parton" with some properties. Instead, within QCD we now know exactly, what the object in question is: a quark, which has a strict definition in QFT with well-defined properties and quantum numbers. For simplicity, we still call an object, which can be either a quark or a gluon "parton".

1.2 Setting the stage

In the following section we collect and define all ingredients needed for Eq. (1.1). As already said, DIS is given by

$$e^-(k) + p(P) \to e^-(k') + X \tag{1.5}$$

FH: copy DIS diagram from PDG where the four-momenta k, P, k' belong to the incoming electron, the incoming proton and the outgoing electron respectively. This allows us, first, to define four momentum of the exchanged photon FH!

$$q = k - k' \tag{1.6}$$

and from there the set of four common variables:

$$s = (k + P)^2 = \frac{Q^2}{xy} \tag{1.7}$$

$$Q^2 = -q^2 \tag{1.8}$$

$$x = \frac{Q^2}{2q \cdot P} \tag{1.9}$$

$$y = \frac{q \cdot P}{k \cdot P} \tag{1.10}$$

We refer to Q^2 as virtuality (of the exchanged boson), to x as Bjorken-x (named after James Bjorken), and to y as inelasticity. Note that Eq. (1.7) gives an explicit relation among these, thus we have only three independent and one dependent variable. In practice, the invariant mass of the electron-proton system s is fixed by the experimental setup, i.e. by the energy of the colliding particles. Also the measurable range of y is in practice limited by the experimental geometry. DIS cross sections are typically provided as double-differential cross sections in x and y. Finally, it is convenient to define the mass squared of the system X recoiling against the scatter electron,

$$W^2 = (P + q)^2 = Q^2 \left(\frac{1}{x} - 1 \right) \tag{1.11}$$

Conclusion 1.5: DIS requirements

We require $Q^2 \gg \Lambda^2_{\text{QCD}}$ (deep) and $W^2 \gg \Lambda^2_{\text{QCD}}$ (inelastic) to call a lepton-hadron scattering DIS.

We can decompose the (double-differential) cross section into a leptonic interaction and a hadronic interaction, yielding each a tensor in Minkowski space, $L_{\mu\nu}$ and $W_{\mu\nu}$ respectively. Thus, we can write

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{2\pi y \alpha^2_{\text{em}}}{Q^4} L_{\mu\nu} W^{\mu\nu} \tag{1.12}$$

where we have introduced the electro-magnetic coupling α_{em}, which appears once on the leptonic side and once on the hadronic side. We can express the leptonic tensor using plain Quantum Electrodynamics (QED) and provide a decomposition of the hadronic tensor in terms of scalar functions. Together, we obtain

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha_{\text{em}}^2}{xyQ^2}\left(\left(1-y+\frac{y^2x}{2}\right)F_2(x,Q^2) - \frac{y^2x}{2}F_L(x,Q^2)\right) \tag{1.13}$$

where we have introduced the two common scalar functions, now called structure functions F_2 and F_L, which are often measured in experiments. It is convenient to also define a third, linearly dependent structure function via

$$2xF_1(x,Q^2) = F_2(x,Q^2) - F_L(x,Q^2)\,. \tag{1.14}$$

Keep in mind that the discussion so far is a major simplification of DIS and in Section 2.1 we reintroduce the main complexity.

Let us continue to collect the remaining ingredient for Eq. (1.1):

- The sum over partons j includes quarks and gluons and we refine in Section 2.5 how large the sum actually is.
- The parton momentum fraction $\xi = x/z$ is the fraction of the four momentum the parton carries with respect to its parent proton.
- The probability of such a configuration to occur is given by the PDF, i.e. $f_j(\xi,Q^2)\mathrm{d}\xi$ is the probability to find a parton j with momentum ξP inside a proton with momentum P at an experiment performed at the scale Q^2.
- The integration over z sums all configuration to generate the final state configuration.

Finally, the coefficient functions $\mathbf{C}(z,Q^2)$ are computable order-by-order in pQCD and thus can be decomposed as

$$\mathbf{C}(z,Q^2) = \sum_{k=0}\alpha_s^k(Q^2)\mathbf{C}^{(k)}(z)\,. \tag{1.15}$$

Note that the Q^2 dependency of $\mathbf{C}(z,Q^2)$ is fully captured by the strong coupling $\alpha_s(Q^2)$.

1.3 Distributions

As we have seen the coefficient functions may contain distributions. For example the quark coefficient function at LO is just a Dirac delta distribution:

$$C_q^{(0)}(z) = \delta(1-z)\,. \tag{1.16}$$

To better compare with our next step, let's repeat the definition of the Dirac delta distribution:

$$\int_0^1 \mathrm{d}z\, \delta(1-z) f(z) = f(1). \tag{1.17}$$

However, this is only the simplest case, which appears at LO, and at higher order we get more complicated distributions. For example the quark coefficient function at NLO is given by

$$\begin{aligned} C_q^{(1)}(z) = C_F \Bigg[& 4\left(\frac{\ln(1-z)}{1-z}\right)_+ - 3\left(\frac{1}{1-z}\right)_+ - (9+4\zeta_2)\delta(1-z) \\ & -2(1+z)\ln\left(\frac{1-z}{z}\right) - 4\frac{\ln(z)}{1-z} + 6 + 4z \Bigg] \end{aligned} \tag{1.18}$$

where the first line only contains distributions. Here, we have introduced the usual $+$-distributions, which are defined via

$$\int_0^1 \mathrm{d}z\, a(z)_+ f(z) = \int_0^1 \mathrm{d}z\, a(z) \left(f(z) - f(1)\right)\,. \tag{1.19}$$

We review in Section 1.4 again briefly on how these distributions appear, but for the moment we turn to more practical matters as the goal of our course is to get an actual comparison to experimental data. The problem is: distributions are (complicate) math objects, defined only via integral relations, Eqs. (1.17) and (1.19), which are unsuitable for numeric evaluation.

1.4 Next-to-leading order

At LO accuracy, DIS is trivial FH: add LO diagram and we find

FH!

$$C_u^{(0)}(z) = e_u^2\delta(1-z) = C_{\bar{u}}^{(0)}(z) = C_c^{(0)}(z) = C_{\bar{c}}^{(0)}(z) = \cdots \tag{1.20}$$
$$C_d^{(0)}(z) = e_d^2\delta(1-z) = C_{\bar{d}}^{(0)}(z) = C_s^{(0)}(z) = C_{\bar{s}}^{(0)}(z) = \cdots \tag{1.21}$$

which, eventually, yields Eq. (1.4).

Starting from next-to-leading order (NLO), the diagrams become more complicated and we encounter new features.

- NLOq
- NLOg

We need the Riemann Zeta function $\zeta(n)$ below, which is defined at integer values n by

$$\zeta_n = \zeta(n) = \sum_{j=1}^{\infty} \frac{1}{j^n} \tag{1.22}$$

Chapter 2

DIS in real life

2.1 DIS is diverse

In this section we give an overview many different DIS datasets commonly used in PDF fits. By doing so, we quickly realize that the picture in Chapter 1 is far too simplistic.

Figure 2.1: "The view into the 6.3-kilometre-long HERA tunnel shows the superconducting magnets for guiding the protons on top and the electron ring beneath them." - taken from [21]

Table 2.1: HERA measurement [22] used in NNPDF4.0 [2]

projectile	$\sqrt{s}$ [GeV]	x	Q^2	$N_{\text{dat}}^{\text{fit}}/N_{\text{dat}}$	obs.
e^-	318	?	?	?/201-43	σ_r^{NC}
e^+	318	?	?	?	σ_r^{NC}
e^+	300	?	?	?	σ_r^{NC}
e^+	251	?	?	?	σ_r^{NC}
e^+	225	?	?	?	σ_r^{NC}
e^-	318	?	?	?	σ_r^{CC}
e^+	318	?	?	?	σ_r^{CC}

2.1.1 HERA

We start with the Hadron-Elektron-Ringanlage (HERA), which was in operation between 1992 and 2007. HERA was hosted by DESY in Hamburg, Germany and Fig. 2.1 shows an image from the HERA tunnel. While the promoted physics discovery, leptoquarks, eventually did not happen, HERA has become one of the most influential DIS machines so far. In fact, it is until this day the only lepton-proton collider in the world. It recorded a huge amount of data, which is still analyzed nowadays, 20 years after the shutdown. HERA had four experiments (H1, ZEUS, HERA-B and HERMES), but we focus here on the combined analysis of H1 and ZEUS [22]. In Table 2.1 we given an overview of the most important fully-inclusive DIS measurements from HERA.

FH!

FH: complete HERA table

We start by looking at the projectile column and immediately we realize that HERA was most of the time not colliding electrons, but positrons. This is to remind us that DIS is lepton-hadron scattering and, indeed, we have to consider all leptons. We expand our selection further below in Section 2.1.3, but remain for the moment with the electron and the positron, which are exact copies of each other, but with opposite charge.

Next, we turn to the energy column $\sqrt{s}$, which spreads across four different energies. In practice the lepton beam was operated with a fixed beam energy ($E_e = 27.5\,\text{GeV}$) and varying energies for the protons ($E_p = 920\,\text{GeV},\ 820\,\text{GeV},\ 575\,\text{GeV}$ and $460\,\text{GeV}$). If we then look to the x and Q^2 columns an recall Eq. (1.7), we see that the table only indicates the maximal available range of those variables. Specifically, HERA was able to measure activity in the detector for $0.005 < y < 0.95$.

However, not only the accessible kinematical phase space limits our kinematic coverage, but also our requirement to study DIS, conclusion 1.5. In practice, e.g. NNPDF4.0 [2] applies a cut of $Q^2 \geq 8\,\mathrm{GeV}^2$ and $W^2 \geq 12.5\,\mathrm{GeV}^2$. Mathematically, the Q^2 cut ensures we are in a perturbative regime, Eq. (1.15), while the W^2 cut has multiple motivations. First, it ensures we are in an inelastic regime, i.e. we are not finding resonances or excitations of the proton. Second and related to the former, we know it suppresses the correction term $\mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right)$ in Eq. (1.1). Studying and trying to estimate the correction term is a topic in current research [23–25]. Looking to the column with the number of data points, we note that even after the kinematic cuts we are left with a total of FH: $N^{\mathrm{HERA}}_{\mathrm{data}}$ measured cross sections. FH!

Finally, we turn to the observables column for which we see two different entries exist: the reduced neutral current (NC) cross section σ_r^{NC} and the reduced charge current (CC) cross section σ_r^{CC}. The "current" here refers to the current between the leptonic part and the hadronic part, in particular the exchanged boson. The "reduced" cross section refers to a slight rescaling of the double differential cross section, Eq. (1.13),

$$\sigma_r = \frac{xyQ^2}{4\pi\alpha^2_{\mathrm{em}}}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} \tag{2.1}$$

such that it refers to a specific PDF combination at LO accuracy.

Next, we examine the CC case by recalling two important facts from conclusion 1.1: first, charged leptons, and thus also electrons and positrons, have weak charges and, second, W bosons have an electric charge. Thus, by CC we consider the lepton interacts with the hadron via the exchange of a W boson, which behaves quite different then a photon. First and foremost, due to the electric charge conservation the W boson can only interact with specific leptons and quarks. On the leptonic side this implies that, first, e.g., the positron always emits a W^+ boson, and, second, the recoiling lepton is a neutrino, which in practice is impossible to track. Specifically, it is impossible to measure it's momentum four-vector k', Eq. (1.5), and thus measure the W boson momentum q, Eq. (1.6), or any other derived variable (x, Q^2, y). Instead, the full reconstruction of the hadronic final state also allows to extract the relevant kinematic variables, which is used instead [22]. Although the production of CC events is suppressed by the propagator of the W boson, $\sim (Q^2 + m_W^2)^{-1}$, it is a LO effect and thus no further suppression. Moreover, since the W are maximally parity violating this allows a third, independent structure function (typically referred to as F_3) to appear in the decomposition of the cross section, Eq. (1.13). We review the problems introduced by γ_5, which causes the parity violation, further in Section 2.7.

We now consider the NC case, which is a generalization of the photon exchange discussed in Chapter 1. As we recall from conclusion 1.1 the Z boson is practically a massive copy of the photon γ and they share all other quantum numbers. This implies whenever a photon is exchanged, also a Z boson can be exchanged, with the main difference that its contributions are suppressed by the respective propagator, $\sim (Q^2 + m_Z^2)^{-1}$. More specifically, we get three contributions to our final cross section: a pure photon contribution σ^γ (where the superscript γ is often suppressed), a pure Z contribution σ^Z, and an interference contribution[1] $\sigma^{\gamma Z}$, which have to be summed together:

$$\sigma^{\mathrm{NC}} = \sigma^\gamma + \sigma^{\gamma Z} + \sigma^Z\,. \tag{2.2}$$

While at low virtualities $Q^2 \ll m_Z^2$ the Z contributions can be neglected, this does not apply to the largest HERA measurements. Similar to W boson, also Z bosons have an axial-vectorial coupling to fermions, $\sim \gamma^\mu\gamma^5$, which introduces a contribution from F_3 to Eq. (1.13). The interference contribution, $F_{1,2}^{\gamma Z}$, yields an electric charge (from the photon) and a vectorial coupling (from the Z boson) both on the leptonic and the hadronic side, whereas the pure Z contribution, $F_{1,2}^Z$, yields squares of both vectorial and axial-vectorial couplings. On the other side, as F_3 is per-se a parity-violating observable, there the interference contribution $F_3^{\gamma Z}$ yields an electric charge (from the photon) and an axial-vectorial coupling (from the Z) and the pure Z contribution, F_3^Z yields a product of the vectorial and axial-vectorial couplings.

The fact that NC and CC couple completely different to fermions, i.e. quarks and leptons, and the definition of three independent structure functions in both cases allows one to fully determine PDFs from DIS [26]. FH: Point to LO DIS exercise?

FH!

Let us return to the key point of this section here. Based on the discussion above, we can now better justify why we don't consider all particles in Fig. 1.1. According on our definition of DIS, conclusion 1.3, in theory also the Higgs boson would contribute to NC cross section, but since its coupling is proportional to the mass of the particle and we always[2] consider massless leptons it does in practice not contribute. As the tau has a very short lifetime they are in practice impossible to accelerate in a collider and

[1] the interference term typically contains the required factor of 2 from the binomial formula inside

[2] almost always [27] - but even so Higgs would only contribute a very large Q^2

thus not suitable for DIS collider experiments. Instead, while muons are also difficult to accelerate concrete experiments for muon-proton scattering have been proposed [28].

2.1.2 Nuclear DIS

After having considered all available bosons and a few more leptons, we now need to consider more hadrons. While we focus here on the proton PDF point-of-view, DIS is equally important for nuclear PDFs [29–32], which describe the internal structure of nuclei. Moreover, the two are typically directly intertwined: proton PDF extractions often contain measurements on nuclear target [2], while nuclear PDF extraction often require a proton baseline.

2.1.3 Neutrino DIS

- CHORUS, Sami
- FASER
- astro

2.2 Interpolation grids

Since PDFs are non-perturbative objects, encoding low-scale physics, they are not computable from first principles inside pQCD. Instead, PDFs are obtained from a fitting procedure, which can be summarized as follows:

Conclusion 2.1: PDF fitting

1. Guess a candidate PDF
2. Compute the theory predictions with this candidate PDF
3. Compare the theory predictions to the experimental measurement

Repeat the above steps until the procedure converges, i.e. the comparison is “good enough”.

The specific details on how to guess a candidate PDF, how to exactly compute the theory predictions, which experimental data to use, and what "good enough" exactly means is a choice. That is the reason why competing PDF fitting groups exist and why they do not necessarily obtain the same PDF, although we all assume the latter to be universal.

The main point for us here is that we need to compute theory predictions very many times for slightly different PDFs. This is a serious problem for complicated hadronic observables, such as, e.g., jet cross sections [33], where the computation can take many hours. The solution is to compute fast interpolation grids, using, e.g., PineAPPL [34,35], APPLgrid [36], or fastNLO [37], which store the partonic matrix elements in an efficient way, such that any subsequent convolution with an arbitrary PDF is fast.

In DIS this is not a very big problem, since typically coefficient functions, Eq. (1.1), are known analytically and thus the convolution with PDFs is straightforward. However, PineAPPL allows also DIS cross sections to be stored thus we can use this simplified case to introduce the general concept here. Moreover, having all observables in a PDF fit sharing the same technology also simplifies their consistent treatment and inside the NNPDF collaboration this is realized in the pineline framework [38].

Interpolation grids are only a discretized version of the coefficient functions which we can recombine in a suitable way a posteriori. Specifically, they are higher dimensional tensors on which we perform fast linear algebra operations. The first three major dimensions are:

bin This is the list of cross section that we need to compute simultaneously. In DIS this typically corresponds to all N_{dat} data point measured in a given configuration, e.g. in Table 2.1. Thus, in practice the actual list is given by the experimental measurement.

order This is the list of perturbative orders, i.e. we store each $\mathbf{C}^{(k)}(z)$, Eq. (1.15), separately. Having this list available allows to use a given grid also at lower orders and thus study, e.g., perturbative stability. In practice, the actual list is given by the requested accuracy of the user (provided we can actual fulfill that request). Recall that LO, Eq. (1.16), and NLO, e.g. Eq. (1.18), are quite different and this generates correlations with the momentum fraction dimension (see below).

channel This is the list of PDF flavor combinations, i.e. the sum over j in Eq. (1.1). Having this list available allows to study the impact of different PDF combination and, e.g., determine which PDF the observable can potentially constrain most. In DIS it is given by the coupling structure of the exchanged boson, i.e. γ/Z vs. $W^\pm$. In practice, the actual list is the easiest given directly in terms of individual parton flavors as the available combinations are too complex (see Section 2.4 for additional complications).

These three dimensions are present in any interpolation grid and in DIS there are two additional dimensions:

scale This is the list of factorization and renormalization scales, discussed in Section 2.4. In DIS this dimension turns out to be trivial, i.e. it has support by a single value, as in fully inclusive DIS there is only a single scale, Q^2, available and we set typically $\mu_F^2 = \mu_R^2 = Q^2$. In practice, the actual list is given by the Q^2 specified in the bin configuration.

momentum fraction This is the list of momentum fractions z, Eq. (1.1), over which the coefficient functions are convolved with the PDF. In DIS we are facing the problem of distributions, see Section 1.3, i.e. we can not directly tabulate the coefficient functions themselves. In practice, the actual list is given by a convolution with a suitable set of interpolation functions, which form a basis for PDFs.

For fully inclusive DIS we end up with five dimensional tensors, where each inner dimension may depend on the outer. FH!

FH: explain interpolation?

2.3 Resummation

In preparation to Section 2.4 we need to review two main ingredients into pQCD: the strong coupling $\alpha_s(Q^2)$, Eq. (1.15), and the PDFs $\mathbf{f}(\xi, Q^2)$, Eq. (1.1). Specifically, we need to remind ourselves on their scale dependency if we want to vary them in Section 2.4. While the two objects obey very similar equations, the strong coupling $\alpha_s(Q^2)$ is conceptually simpler and we start with that.

2.3.1 The strong coupling

The strong coupling α_s is directly linked to the coupling of the gluons to quarks,

$$\mathcal{L}_{\mathrm{QCD}} \supset g T_a \bar{\psi} \gamma^\mu A^a_\mu \psi, \qquad \alpha_s = \frac{g^2}{4\pi} \tag{2.3}$$

and is the central object in pQCD around which we organize our perturbative series. If one considers higher order corrections to the gluon-quark couplings, one encounters quickly divergencies of ultra-violet origin, i.e. for large loop momenta. This forces the introduction of a regularization procedure, typically dimensional regularization, which eventually leads to a scale dependent definition of the renormalized strong coupling $\alpha_s(\mu_R^2)$. The introduced scale, called renormalization scale μ_R henceforth, is an arbitrary scale, which is an artifact of our regularization procedure and which effectively represents a separation between perturbative and non-perturbative regimes. However, the scale dependence of $\alpha_s(\mu_R^2)$ is rigorously predicted in pQCD, organizing the terms in such a way that within the requested accuracy any dependence of μ_R is cancelled. Recall that the scale dependence can be computed once the Lagrangian $\mathcal{L}$ is defined and does not require any actual observable.

Eventually, we obtain the renormalization group equation (RGE) of the strong coupling,

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = \beta(\alpha_s) = -\sum_{j=0} \beta_j \left(\alpha_s(\mu_R^2)\right)^{2+j} \tag{2.4}$$

which is also the definition[3] of the beta function $\beta(\alpha_s)$. The upper limit of the sum in Eq. (2.4) defines the perturbative accuracy, to which all other objects in pQCD refer to.

At LO accuracy we find

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = -\beta_0 \left(\alpha_s(\mu_R^2)\right)^2, \qquad \beta_0 = \frac{11 C_A - 4 T_R n_f}{12\pi} \tag{2.5}$$

where we used the second Casimir constant of the adjoint representation $C_A = N_c = 3$, the normalization of the fundamental generators $T_R = 1/2$, and the number of active

[3] Note that in the literature many conventions for the left- and/or right-hand-side exist.

flavors n_f, discussed in Section 2.5. In this specific case we can solve Eq. (2.5) and obtain

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \frac{\alpha_s(\mu_{R,0}^2)}{1 + \alpha_s(\mu_{R,0}^2)\beta_0 \ln\left(\mu_R^2/\mu_{R,0}^2\right)} \tag{2.6}$$

for a given boundary condition $\alpha_s(\mu_{R,0}^2)$ at the scale $\mu_{R,0}^2$. Recall, that Eq. (2.6) has not two free parameters, $\mu_{R,0}$ and $\alpha_s(\mu_{R,0})$ as one would naively think, but only one. FH: excercise: proof lambda in alphas? FH!

Now, coming back to the title of this section, note that Eq. (2.6) is a resummation:

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \alpha_s(\mu_{R,0}^2) \sum_{k=0}^{\infty} \left(-\alpha_s(\mu_{R,0}^2)\beta_0 \ln\left(\mu_R^2/\mu_{R,0}^2\right)\right)^k \tag{2.7}$$

i.e. it collects an infinite tower of logarithms, here $\ln\left(\mu_{R,0}^2/\mu_R^2\right)$, into a compact expression, Eq. (2.6). Since $\mu_{R,0}$ is an arbitrary scale, often chosen to coincide with the mass of the Z boson $\mu_{R,0} = m_Z$, it is basically always required to perform the resummation and use Eq. (2.6). However, in some situations, to be discussed in Section 2.4, we need Eq. (2.6) truncated to a finite order, i.e. unraveling Eq. (2.7), and we get

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \alpha_s(\mu_{R,0}^2) - \left(\alpha_s(\mu_{R,0}^2)\right)^2 \beta_0 \ln\left(\mu_R^2/\mu_{R,0}^2\right) + \dots \tag{2.8}$$

While for the special case $\mu_{R,0} = \mu_R$ we need to be consistent of course and we thus find that $\alpha_s^{\mathrm{LO}}(\mu_R^2)$ is $\alpha_s(\mu_{R,0}^2)$ at the LO truncation, we note that they differ by higher order terms. Equation (2.8) is the first, simplest case of a perturbative expansion of a resummed expression, Eq. (2.6), and can also be obtained by using the Taylor expansion of $\alpha_s(\mu_R^2)$ around $\mu_{R,0}^2$ together with Eq. (2.5). In other words, Eq. (2.8) is a perturbative expansion of Eq. (2.6) in powers of $\alpha_s(\mu_{R,0}^2)$. Note that Eq. (2.7) has a finite radius of convergence (as all geometric series) and that is also reflected in Eq. (2.6).

Starting from NLO accuracy the RGE, e.g. in this specific case given by

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = -\beta_0\left(\alpha_s(\mu_R^2)\right)^2 - \beta_1\left(\alpha_s(\mu_R^2)\right)^3, \tag{2.9}$$

can no longer be solved exactly with elementary functions. However, it is still possible to find an analytic solution which solves Eq. (2.9) at the requested accuracy, i.e. in this case NLO. FH: exercise: proof NLO alphas? This solution still performs a resummation, as we have to, but it only collects the so-called leading logarithms FH: show exp alphas FH! FH!

is LL A completely different way to solve Eq. (2.9) is using numerical tools instead. We refer to the literature on how to implement a specific solution in practice and remain here just with the fact that it is a complicated problem on its own, which has no unique solution and for which approximations are mandatory.

Conclusion 2.2: Solution of RGE

- The LO RGE of the strong coupling, Eq. (2.5), can be solved exactly, Eq. (2.6)
- The RGE of the strong coupling at higher orders, e.g. Eq. (2.9), can not be solved exactly and one needs to choose a solution method

Any proper solution of the RGE must solve the RGE itself at the requested accuracy else it is not a solution, but they can and will differ by terms beyond - we refer to the uncertainty which is associated with this freedom as Missing Higher Order Uncertainty (MHOU). Truncating the perturbative series is a fundamental part of pQCD and thus we are always bound to the accuracy which we request at the beginning. While in many simple application MHOU are often neglected they are an important ingredient in modern PDF fits [39] and in Section 2.4 we demonstrate one possible way to estimate them.

2.3.2 DGLAP

For PDFs $\mathbf{f}(\xi, \mu_F^2)$ the situation is essentially the same as for the strong coupling $\alpha_s(\mu_R^2)$, except the latter is just a number and the former are a set of functions. The scale dependence is a again predicted by pQCD, but now we need an arbitrary physical observable to do so. The specific observable does not matter since PDFs are universal, but due to the linear dependence on PDF and the “simplicity” of fully inclusive DIS it often serves as reference case. The arbitrary scale that is introduced in the regularization procedure, typically called collinear factorization, is referred to as factorization scale μ_F henceforth.

Eventually, we obtain the RGE of PDFs, often referred to as DGLAP equation named after the researcher who established it [40–42],

$$\mu_F^2 \frac{\mathrm{d}\mathbf{f}(\xi, \mu_F^2)}{\mathrm{d}\mu_F^2} = \left(\mathbf{P}(\mu_F^2) \otimes \mathbf{f}(\mu_F^2)\right)(\xi) = \sum_{j=0} \left(\alpha_s(\mu_F^2)\right)^{1+j} \Big(\mathbf{P}^{(j)} \otimes \mathbf{f}(\mu_F^2)\Big)(\xi) \tag{2.10}$$

where $\mathbf{P}(y,\mu_F^2)$ is the matrix of Altarelli-Parisi splitting functions. Note that the upper limit of the sum in Eq. (2.10) must be set to the requested perturbative accuracy, i.e. consistent with Eq. (2.5).

When attempting to solve Eq. (2.10) we must worry about the two complications in PDFs with respect of the strong coupling. First, since PDFs are a vector and thus the Altarelli-Parisi splitting functions are combined in a matrix $\mathbf{P}$, we need to worry about non-commutativity. Indeed, while at LO accuracy we only have a single matrix $\mathbf{P}^{(0)}$, which trivially commutates with itself, starting from NLO we encounter a second matrix $\mathbf{P}^{(1)}$, which does not commutate with the former. Second, at any fixed scale PDFs are a function of the momentum fraction ξ and, moreover, the right-hand-side of Eq. (2.10) features an integral (with the definition of the convolution $\otimes$ in Eqs. (1.1) and (1.2)).

To overcome the second problem, one possibility is to apply an integral transformation, the so-called Mellin transformation, defined by

$$\tilde{\mathbf{f}}(N,\mu_F^2) = \int_0^1 \mathrm{d}\xi\, \xi^{N-1}\mathbf{f}(\xi,\mu_F^2) \tag{2.11}$$

$$\gamma(N,\mu_F^2) = \int_0^1 \mathrm{d}y\, y^{N-1}\mathbf{P}(y,\mu_F^2) \tag{2.12}$$

where the Mellin-transformed splitting functions γ are often referred to as anomalous dimensions. Inserting Eqs. (2.11) and (2.12) into Eq. (2.10) yields a “simple” first order differential equation, which can be solved by a “simple” exponentiation. However, in practice PDFs are mostly needed in momentum fraction space, $\mathbf{f}(\xi,\mu_F^2)$, and not in Mellin space, $\tilde{\mathbf{f}}(N,\mu_F^2)$, thus if we use the Mellin transform, we must also use its inverse transformation.

While we skip spelling out an explicit solution and refer to the literature instead [43], the important part here is that the exponential again captures a resummation, in this case of $\ln\left(\mu_F^2/\mu_{F,0}^2\right)$. Again, we can perform a perturbative expansion of the LO resummed solution to see the logarithms explicitly:

$$\mathbf{f}^{\mathrm{LO}}(\xi,\mu_F^2) = \mathbf{f}(\xi,\mu_{F,0}^2) + \alpha_s(\mu_{F,0}^2)\ln\left(\mu_F^2/\mu_{F,0}^2\right)\left(\mathbf{P}^{(0)}\otimes\mathbf{f}(\mu_{F,0}^2)\right)(\xi) + \ldots \tag{2.13}$$

Also, starting from NLO accuracy, we face the same issues as for the strong coupling and conclusion 2.2 applies in the same way.

2.4 Scale variations

We now attempt to address a problem we first see in Section 2.3: Missing Higher Order Uncertainty (MHOU). We repeat again, the truncation of the perturbative series is a fundamental part of our framework and we are bound to it. The naive solution of just computing the next higher correction is not a solution, as we would face the problem just again there. Thus the task is to get an estimate of the missing terms based on current knowledge, which by definition is an ill-posed problem. Several possible strategies have been suggested in the literature [39, 44–50], which are based on resummation of specific variables and/or statistical arguments.

From the practical point of view as a PDF fitter, however, there is one big problem with many of the proposed algorithms: they only work for some specific set of observables and a detailed analysis of the kinematics. Thus, we review here the most common approach to estimate MHOU, which can be rigorously derived in pQCD and applied to any observable: scale variations. On the other side, especially in recent years, scale variations have been challenged as suitable tool, due to the inherent arbitrariness and the lack of statistical interpretation (to which we return in the end). For now we focus on their practical implementation in DIS.

2.4.1 Renormalization scale

We start again from the simpler case of the renormalization scale μ_R associated with the strong coupling $\alpha_s(\mu_R^2)$ via the RGE, Eq. (2.5). As we say above, the renormalization scale captures the ultra-violet divergencies, which arise, e.g., from self-energy diagrams and which are thus associated with the partonic diagram itself. Moreover, as we see from Eq. (1.15), we expand the partonic coefficient functions in powers of the strong coupling.

Conclusion 2.3: Renormalization scale dependence

The renormalization scale dependence is captured by the partonic coefficient function $\mathbf{C}(z, Q^2, \mu_R^2)$.

Since we have to choose something for the arbitrary scale μ_R and we only have a single scale available, the virtuality Q^2, we start by setting them equal to each other $\mu_R^2 = Q^2$, which is also the implicit choice in Eq. (1.1). We then need to find a way to estimate the missing terms in $\mathbf{C}(z, Q^2, \mu_R^2 = Q^2)$, i.e. something that is identical at the requested perturbative accuracy, but differs from it beyond. This can be achieved by using Eq. (2.8) together with a suitable redefinition of the perturbative coefficients of the partonic coefficient function $\mathbf{C}^{(j)}(z)$. Specifically, we require

$$\mathbf{C}(z, Q^2, \mu_R^2) = \mathbf{C}(z, Q^2, \mu_R^2 = Q^2)\big(1 + \alpha_s(Q^2)\big) \tag{2.14}$$

with

$$\mathbf{C}(z, Q^2, \mu_R^2) = \sum_{j=0} \big(\alpha_s(\mu_R^2)\big)^j \mathbf{C}^{(j)}(z, \ln\big(\mu_R^2/Q^2\big))\,. \tag{2.15}$$

Before we continue with deriving explicit expressions for $\mathbf{C}^{(j)}(z, \ln\big(\mu_R^2/Q^2 0\big))$ some comments are in order:

- $\mathbf{C}(z, Q^2)$ in Eq. (1.1) is an abbreviation of $\mathbf{C}(z, Q^2, \mu_R^2 = Q^2)$
- in Eq. (2.15) μ_R has its intended meaning: it is the scale of the strong coupling
- from Section 2.3.1 we know $\alpha_s(\mu_R^2)$ is a pure resummation of logarithms, thus $\mathbf{C}^{(j)}$ may only depend on $\ln\big(\mu_R^2/Q^2\big)$
- we also know that it must be a polynomial dependence since it is a pure resummation

2.5 Heavy Quarks

- F_2^c
- FONLL

2.6 TMC

2.7 γ_5

- helicity
- HVBM + Larin

Chapter 3

More advanced topics

3.1 Positivity

- DIS vs. MSbar

3.2 Higher orders

- IHOU
- N3LO
- channels
- loops and legs

3.3 QED

- Qiu et al.

3.4 Valo

- $C_\gamma \to k$

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